

Assignment 2A - One-Time Pad Encryption and Decryption

Consider a character set having 64 characters as follows. There are 26 uppercase letters (A ... Z), 26 lowercase letters (a ... z), 10 digits (0 ... 9), and the two special characters (+ and -). The given table in page 2 shows the character set and the 6-bit code for every character.

Requirements and Deliverables:

- 1- Use a one-time pad to encrypt and decrypt the text message {**Hello World**}. Each student **must** generate his own random one-time pad key using letters obtained from his name as shown next in the solution steps.
- 2- Print and deliver a single-page report (as shown in page 3) showing all computation steps for encryption and decryption that are detailed below.

Solution Steps:

- 1- To generate the random key, write down the first 26 characters of your full name as shown in the table below and select the 10 characters marked by '*'. Do **not** change the positions of the '*' markings.
- 2- Here is an example in which the marked characters are {M, a, e, m, l, b, e, H, s, n}.

M	o	h	a	m	e	d	A	h	m	e	d	l	b	r	a	h	e	e	m	H	a	s	s	a	n		
*			*		*				*			*	*				*			*			*		*		

- 3- Use the given table to get the 6-bit code for every marked character. These are 10 groups of 6-bits, The one-time pad key is the 60 bits corresponding to the 10 marked characters in the given sequence.
- 4- Convert the plaintext "**Hello World**" into the corresponding bits using the given table.
- 5- Do the encryption by XORing the plaintext with the key bit by bit.
- 6- Find the ciphertext by replacing every 6 bits of the XOR output with their corresponding character from the table.
- 7- For decryption, repeat steps 5 and 6 by XORing the **same key** with the obtained ciphertext. The decrypted ciphertext **must be** the same as the plaintext.

Table for the binary values of the available 64-character set

Nr.	Ch.	6-bits code							Nr.	Ch.	6-bits code					
0	A	0	0	0	0	0	0		32	g	1	0	0	0	0	0
1	B	0	0	0	0	0	1		33	h	1	0	0	0	0	1
2	C	0	0	0	0	1	0		34	i	1	0	0	0	1	0
3	D	0	0	0	0	1	1		35	j	1	0	0	0	1	1
4	E	0	0	0	1	0	0		36	k	1	0	0	1	0	0
5	F	0	0	0	1	0	1		37	l	1	0	0	1	0	1
6	G	0	0	0	1	1	0		38	m	1	0	0	1	1	0
7	H	0	0	0	1	1	1		39	n	1	0	0	1	1	1
8	I	0	0	1	0	0	0		40	o	1	0	1	0	0	0
9	J	0	0	1	0	0	1		41	p	1	0	1	0	0	1
10	K	0	0	1	0	1	0		42	q	1	0	1	0	1	0
11	L	0	0	1	0	1	1		43	r	1	0	1	0	1	1
12	M	0	0	1	1	0	0		44	s	1	0	1	1	0	0
13	N	0	0	1	1	0	1		45	t	1	0	1	1	0	1
14	O	0	0	1	1	1	0		46	u	1	0	1	1	1	0
15	P	0	0	1	1	1	1		47	v	1	0	1	1	1	1
16	Q	0	1	0	0	0	0		48	w	1	1	0	0	0	0
17	R	0	1	0	0	0	1		49	x	1	1	0	0	0	1
18	S	0	1	0	0	1	0		50	y	1	1	0	0	1	0
19	T	0	1	0	0	1	1		51	z	1	1	0	0	1	1
20	U	0	1	0	1	0	0		52	0	1	1	0	1	0	0
21	V	0	1	0	1	0	1		53	1	1	1	0	1	0	1
22	W	0	1	0	1	1	0		54	2	1	1	0	1	1	0
23	X	0	1	0	1	1	1		55	3	1	1	0	1	1	1
24	Y	0	1	1	0	0	0		56	4	1	1	1	0	0	0
25	Z	0	1	1	0	0	1		57	5	1	1	1	0	0	1
26	a	0	1	1	0	1	0		58	6	1	1	1	0	1	0
27	b	0	1	1	0	1	1		59	7	1	1	1	0	1	1
28	c	0	1	1	1	0	0		60	8	1	1	1	1	0	0
29	d	0	1	1	1	0	1		61	9	1	1	1	1	0	1
30	e	0	1	1	1	1	0		62	+	1	1	1	1	1	0
31	f	0	1	1	1	1	1		63	-	1	1	1	1	1	1

Solution Report for Assignment – 2A

St.Name: Hamza Mohammad Ismail Algohary

St.No: 27

Write your full name in English below. The first letters of the words must be uppercase, and the other letters are lowercase. Do not use spaces. Select the characters marked by ''.*

H	a	m	z	a	M	o	h	a	m	m	a	d	l	s	m	a	i	l	A	l	g	o	h	a	r	y	
*			*		*				*			*	*			*			*			*		*	*		

The selected characters are:

HzMmdlilhr

The one-time pad key is:

000111 , 111001 , 001100 , 101100 , 100011 , 001000 , 101000 , 101011 , 100111 ,
110001

The plaintext bits are:

HelloWorld

The ciphertext bits are:

000000 , 101101 , 101001 , 000011 , 110101 , 011110 , 001010 , 001110 , 000100 ,
110110

The ciphertext characters are:

AtpD1eKOE2

The decrypted ciphertext bits are:

000111 , 011110 , 100101 , 100101 , 101000 , 010110 , 101000 , 101011 , 100101 ,
011101

The decrypted ciphertext characters are:

HelloWorld